1. AWS Storage Services — 30 Important Interview Questions & Answers

1. What are AWS Storage Services?
Answer:
AWS provides different storage services for different types of data and application needs. The main storage services include Amazon S3 for object storage, Amazon EBS for block storage, and Amazon EFS for shared file storage. Each service is designed for a different use case.
For example, an EC2 server can use EBS for its disk, multiple servers can use EFS for shared files, and application files or images can be stored in S3.
Example:
AWS Storage
|

| | |
S3 EBS EFS
Object Block File
Simple ga gurthupettuko:
S3 = Object
EBS = Block
EFS = File

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2. What is Amazon S3?
Answer:
Amazon S3 stands for Simple Storage Service. It is an object storage service used to store and retrieve files and other data. S3 is highly scalable and is commonly used for images, videos, backups, logs, documents, and application files.
For example, a website can store user profile pictures in an S3 bucket instead of storing them on an EC2 server.
Example:
Website
↓
S3 Bucket
↓
Images / Videos / Files
Simple ga gurthupettuko:
S3 = Object Storage

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3. What is an S3 Bucket?
Answer:
An S3 bucket is a container used to store objects in Amazon S3. Objects can be files, images, videos, documents, backups, and other data. Each bucket has a unique name within the AWS S3 namespace.
For example, we can create a bucket called my-company-images and store user profile pictures inside it.
Example:
S3 Bucket
↓
---
image1.jpg
image2.jpg
video.mp4
document.pdf
Simple ga gurthupettuko:
Bucket = Container for S3 objects

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4. What is an S3 Object?
Answer:
An object is the actual data stored inside an S3 bucket. An object contains the file data and information such as its key and metadata. The key identifies the object inside the bucket.
For example, profile/ravi.jpg can be the key of a profile image stored in an S3 bucket.
Example:
Bucket
↓
profile/ravi.jpg
↓
Image Data
Simple ga gurthupettuko:
Object = Actual file/data in S3

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5. What is Amazon EBS?
Answer:
Amazon EBS stands for Elastic Block Store. It provides block storage volumes that can be attached to EC2 instances. EBS is commonly used as the disk storage for EC2 servers.
For example, an EC2 server running Linux can use an EBS volume to store its operating system, application files, and other data.
Example:

EC2
↓
EBS Volume
↓
Application Data
Simple ga gurthupettuko:
EBS = Disk for EC2

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6. What is Amazon EFS?
Answer:
Amazon EFS stands for Elastic File System. It provides shared file storage that can be accessed by multiple EC2 instances. It is useful when multiple servers need to read and write the same files.
For example, three EC2 instances can use the same EFS file system to access shared application files.
Example:
EC2-1 ─┐
EC2-2 ─┼─ EFS
EC2-3 ─┘
Simple ga gurthupettuko:
EFS = Shared File Storage

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7. What is the difference between S3, EBS, and EFS?
Answer:
S3 is object storage, EBS is block storage, and EFS is file storage. S3 is mainly used for files and objects, EBS is commonly used as storage for EC2, and EFS is used when multiple servers need shared file access.
For example:
S3 → Images / Videos / Backups

EBS → EC2 Disk

EFS → Shared Files
Simple ga gurthupettuko:
S3 = Object
EBS = Block
EFS = File

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8. Which AWS storage should we use for EC2 root disk?
Answer:
Amazon EBS is commonly used for the root disk of an EC2 instance. It provides block storage that the operating system can use like a normal disk. The root EBS volume is created based on the AMI and launch configuration.
For example:
AMI
↓
Root EBS Volume
↓
EC2
Simple ga gurthupettuko:
EC2 Root Disk = EBS

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9. Which storage should we use for images and videos?
Answer:
Amazon S3 is commonly used for images, videos, documents, backups, and other files. It is designed for object storage and can store very large amounts of data. Applications can access these objects when needed.
For example, a social media application can store user-uploaded photos and videos in S3.
Example:
User
↓
Application
↓
S3
↓
Images / Videos
Simple ga gurthupettuko:
Images/Videos/Files = S3

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10. Which storage should we use when multiple EC2 servers need the same files?
Answer:
Amazon EFS is a common choice when multiple EC2 instances need shared file access. The same file system can be mounted by multiple servers. This avoids keeping separate copies of the same files on each server.
For example:
EC2-1 ─┐
EC2-2 ─┼─ EFS

EC2-3 ──┘
Simple ga gurthupettuko:
Multiple EC2 + Same Files = EFS

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11. Is S3 attached directly to EC2 like EBS?
Answer:
No. S3 is object storage and is accessed through APIs or supported interfaces, while EBS is a block storage volume attached to an EC2 instance. They work in different ways.
For example:
EC2 ──attached──> EBS

EC2 ──API/access──> S3
Simple ga gurthupettuko:
EBS = Attached disk
S3 = Object service

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12. Can EBS be attached to multiple EC2 instances?
Answer:
Normally, an EBS volume is attached to one EC2 instance at a time, but certain EBS volume types support multi-attach with supported instance and configuration requirements. We should not assume that every EBS volume can be shared between servers.
For example:
EC2-1
↓
EBS
For supported Multi-Attach scenarios:
EC2-1 ──┐
├── EBS
EC2-2 ──┘
Simple ga gurthupettuko:
EBS sharing depends on volume type and configuration

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13. What is S3 Versioning?
Answer:
S3 Versioning keeps different versions of an object when the object is changed or deleted. It helps protect data from accidental deletion or unwanted changes. We can restore an older version when required.
For example:
file.txt
↓
Version 1
Version 2
Version 3
Simple ga gurthupettuko:
S3 Versioning = Keep old object versions

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14. What is S3 Lifecycle?
Answer:
S3 Lifecycle rules automatically move or delete objects based on rules we define. This helps reduce storage cost and manage old data automatically. For example, old files can move to a cheaper storage class.
For example:
New File
↓
S3 Standard
↓
Old File
↓
Cheaper Storage Class
↓
Delete Later
Simple ga gurthupettuko:
Lifecycle = Automatically move/delete old data

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15. What are S3 Storage Classes?
Answer:
S3 provides different storage classes based on how often data is accessed and how much we want to pay for storage and retrieval. Common classes include S3 Standard, Standard-IA, One Zone-IA, Intelligent-Tiering, Glacier Instant Retrieval, Glacier Flexible Retrieval, and Glacier Deep Archive.
For example, frequently accessed files can stay in S3 Standard, while long-term archive data can use Glacier storage classes.
Example:
Frequently Used → S3 Standard

Rarely Used → IA

Archive → Glacier
Simple ga gurthupettuko:
Storage Class = Choose storage based on data usage

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16. What is S3 Encryption?
Answer:
S3 encryption protects objects stored in S3. We can use server-side encryption options, including S3-managed keys or AWS KMS keys depending on the requirement. Encryption helps protect stored data.
For example:
Application
↓
S3
↓
Encrypted Object
↓
Storage
Simple ga gurthupettuko:
S3 Encryption = Protect stored objects

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17. What is an EBS Snapshot?
Answer:
An EBS Snapshot is a point-in-time backup of an EBS volume. It can be used to create a new EBS volume or support recovery. Snapshots are stored in AWS-managed storage rather than acting as directly attached disks.
For example:
EBS Volume
↓
Snapshot
↓
New EBS Volume
Simple ga gurthupettuko:
EBS Snapshot = Backup/copy of EBS volume

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18. What is EBS encryption?
Answer:
EBS encryption protects data stored on EBS volumes and supported related data such as snapshots. AWS KMS keys can be used to manage encryption keys. Encryption can help protect sensitive application and server data.
For example:
EC2
↓
Encrypted EBS
↓
Protected Data
Simple ga gurthupettuko:
EBS Encryption = Protect EBS data

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19. What is EFS encryption?
Answer:
EFS supports encryption of data at rest and encryption of data in transit. Encryption at rest protects stored files, while encryption in transit protects data while it moves between the EC2 client and EFS. AWS KMS can be used for encryption at rest.
For example:
EC2
↓
Encrypted Connection
↓
EFS
↓
Encrypted Files
Simple ga gurthupettuko:
EFS = Encryption at rest + in transit

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20. What is EFS Mount Target?
Answer:
An EFS mount target provides network access to an EFS file system from a VPC subnet. EC2 instances use the mount target to connect to the EFS file system. EFS commonly has mount targets in the Availability Zones where clients need access.
For example:
EC2
↓
Subnet
↓
EFS Mount Target

↓
EFS
Simple ga gurthupettuko:
Mount Target = Network access point to EFS

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21. What is S3 Bucket Policy?
Answer:
An S3 bucket policy is a resource-based policy that controls access to objects and actions in an S3 bucket. It can allow or deny access to specific AWS accounts, roles, services, or other principals. Bucket policies are commonly used to control access to S3 resources.
For example:
User / Role
↓
Bucket Policy
↓
S3 Bucket
↓
Allow / Deny
Simple ga gurthupettuko:
Bucket Policy = Control access to S3 bucket

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22. What is EBS Volume Type?
Answer:
EBS provides different volume types for different performance and cost requirements. Common SSD types include gp3 and io2, while st1 and sc1 are designed for specific HDD-based workloads. We choose the volume type based on application requirements.
For example:
General Application → gp3

High IOPS Database → io2

Large Throughput Workload → st1
Simple ga gurthupettuko:
EBS Volume Type = Choose performance and cost

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23. What is S3 Static Website Hosting?
Answer:
S3 can host static website files such as HTML, CSS, JavaScript, and images. This is useful for websites that do not require a server-side application running on the storage service itself. For production architectures, S3 can also work with services such as CloudFront.
For example:
User
↓
CloudFront / S3
↓
HTML
CSS
JavaScript
Simple ga gurthupettuko:
S3 Static Website = Host static files

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24. What is S3 Replication?
Answer:
S3 Replication automatically copies objects from one S3 bucket to another bucket based on configured replication rules. It can be used for disaster recovery, compliance, or keeping data in another Region. Cross-Region Replication is commonly used to copy data between AWS Regions.
For example:
S3 Mumbai
↓
Replication
↓
S3 Singapore
Simple ga gurthupettuko:
S3 Replication = Automatically copy objects

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25. What is S3 Object Lock?
Answer:
S3 Object Lock helps prevent objects from being deleted or changed for a defined period or under a retention rule. It is useful when data must be kept for compliance or protection against accidental deletion. It is commonly used for important records and backups.
For example:
Important Backup

↓
S3 Object Lock
↓
Cannot be deleted/changed
during retention
Simple ga gurthupettuko:
Object Lock = Protect important objects from deletion/change

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26. What is AWS Storage Gateway?
Answer:
AWS Storage Gateway is a service that connects on-premises applications with AWS storage. It provides different gateway types for file, volume, and tape-based use cases. It helps organizations move or extend storage into AWS without changing every application immediately.

For example:
On-Premises
↓
Storage Gateway
↓
AWS Storage
Simple ga gurthupettuko:
Storage Gateway = Connect on-premises storage with AWS

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27. What is AWS DataSync?
Answer:
AWS DataSync is used to move data between on-premises storage and AWS storage services, or between supported AWS storage locations. It helps automate and speed up large data transfers. It is useful for migrations and regular data movement.

For example:
On-Premises Files
↓
DataSync
↓
S3 / EFS / FSx
Simple ga gurthupettuko:
DataSync = Move data between storage locations

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28. What is Amazon FSx?
Answer:
Amazon FSx provides managed file systems for specific workloads. AWS offers file systems such as FSx for Windows File Server, FSx for Lustre, FSx for NetApp ONTAP, and FSx for OpenZFS. We choose the file system based on application requirements.

For example:
Application
↓
Amazon FSx
↓
Managed File System
Simple ga gurthupettuko:
FSx = Managed file systems for specific workloads

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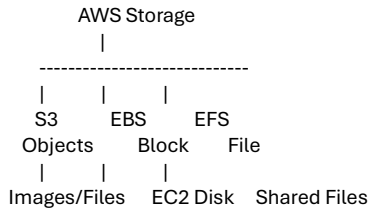
29. How do you choose between S3, EBS, EFS, and FSx?
Answer:
I choose the storage service based on how the application needs to access the data. S3 is for objects, EBS is for block storage attached to compute, EFS is for shared Linux file storage, and FSx is for specific managed file-system requirements.

For example:
Images / Backups → S3
EC2 Disk → EBS
Shared Linux Files → EFS
Windows / Special FS → FSx
Simple ga gurthupettuko:
Choose storage based on **how the application uses the data**.

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30. What is the most important thing to remember about AWS Storage?
Answer:
The most important thing is to understand the difference between object, block, and file storage. S3 is mainly for objects, EBS is mainly for block storage attached to EC2, and EFS is shared file storage. AWS also provides services such as FSx, Storage Gateway, and DataSync for specific storage needs.
For example, if I have an application running on EC2, I can use EBS for the server disk, EFS when multiple servers need the same files, and S3 for images, videos, backups, and other objects.

Example:



Simple ga gurthupettuko:

S3 = Object | EBS = Block | EFS = File | FSx = Managed File System